Exponential Functions & Growth Models

Section 1: Exponential Growth and Decay Fundamentals

1 Which function represents exponential decay with an initial value of 450 and a decay rate of 8% per year?

a. $f(t) = 450(0.92)^t$

b. $f(t) = 450(1.08)^t$

c. $f(t) = 450(0.08)^t$

d. $f(t) = 450 - 0.08t$

2 An exponential function passes through $(0, 6)$ and $(2, 54)$. If $f(x) = ab^x$ where $b > 0$, what is the value of b?

a. 3

b. 9

c. 6

d. $\sqrt{3}$

3 If a quantity grows continuously at an annual rate of 5%, what is the continuous growth model for an initial investment of 1200 dollars?

a. $A(t) = 1200e^{0.05t}$

b. $A(t) = 1200(1.05)^t$

c. $A(t) = 1200(0.05)^t$

d. $A(t) = 1200e^{1.05t}$

Section 2: Doubling Time & Half-Life Calculations

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4 A bacterial culture doubles every 4 hours. If there are initially 500 bacteria, which equation models the population after t hours?

a. $P(t) = 500 \cdot 2^{t/4}$

b. $P(t) = 500 \cdot 2^{4t}$

c. $P(t) = 500 \cdot 4^{t/2}$

d. $P(t) = 500 + 2^{t/4}$

5 A radioactive isotope has a half-life of 15 days. If a sample contains 80 grams initially, how much remains after 45 days?

a. 10 g

b. 20 g

c. 5 g

d. 15 g

6 If the doubling time of an investment is 12 years, what is the annual growth factor b in $A(t) = A_0 b^t$?

a. $2^{1/12}$

b. $12^{1/2}$

c. 2×12

d. $\frac{2}{12}$

Section 3: Equivalent Exponential Forms & Conversions

7 Rewrite $f(t) = 100(1.21)^{t/2}$ in the form $f(t) = 100b^t$. What is b?

a. 1.1

b. 1.21

c. 1.4641

d. 0.605

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8 If $P(t) = 250e^{0.08t}$, what is the equivalent discrete annual growth factor rounded to four decimal places?

a. 1.0833

b. 1.0800

c. 1.0408

d. 1.1664

9 A medication decreases in the bloodstream by 30% every 6 hours. By what factor does it decrease per hour?

a. $(0.70)^{1/6}$

b. $(0.30)^{1/6}$

c. $0.70/6$

d. $(0.70)^{6}$

Section 4: Comparing Linear and Exponential Growth

10 A linear function $L(x) = 50x + 100$ and an exponential function $E(x) = 2^x$ are compared for large $x > 0$. Which statement is true?

a. Eventually $E(x) > L(x)$ for all sufficiently large x

b. Eventually $L(x) > E(x)$ because its initial rate of change is 50

c. They will intersect infinitely many times

d. $L(x)$ and $E(x)$ grow at the same rate asymptotically

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11 Which data set represents an exponential function rather than a linear function?

a. $x: 1, 2, 3, 4$ and $y: 6, 18, 54, 162$

b. $x: 1, 2, 3, 4$ and $y: 6, 12, 18, 24$

c. $x: 1, 2, 3, 4$ and $y: 6, 11, 18, 27$

d. $x: 1, 2, 3, 4$ and $y: 6, 8, 12, 20$

12 What is the domain and range of the function $f(x) = 4 - 3(2)^x$?

a. Domain: $(-\infty, \infty)$; Range: $(-\infty, 4)$

b. Domain: $(0, \infty)$; Range: $(-\infty, 4)$

c. Domain: $(-\infty, \infty)$; Range: $(4, \infty)$

d. Domain: $(-\infty, 4)$; Range: $(-\infty, \infty)$

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